Prob 7. (28), Ch 6.4.1)

The general sol-n for the <u>one-dim-l</u> wave eq-n

(1) $$\boxed{u_{xx} - u_{tt} = 0}$$

is

(2) $$\boxed{u(x,t) = F(x+t) + g(x-t)}$$ D'Alambert sol-n.

where the f-ns F, g are arbitrary f-ns of <u>single</u> variable. This is analogous to the gen-l sol-n for a 2nd order const. coef-t linear ODE which has 2 arbitrary const.s.

(a) Consider the ↓ IVP

(1) $$u_{xx} - u_{tt} = 0$$

(3) $$u(x,0) = \varphi(x) \qquad -\infty < x < \infty$$

(4) $$\frac{\partial u}{\partial t}(x,0) = \psi(x) \qquad -\infty < x < \infty.$$

Using (2) derive the particular sol-n.

(5) $$u(x,t) = \frac{1}{2}\left[\varphi(x+t) + \varphi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \psi(\tau)\, d\tau.$$

Sol-n (a): Using (2), note the ↓ partials & IV.s

(6) $$u_t = F'(x+t) - g'(x-t)$$ &

$$\begin{cases} u(x,0) = F(x+0) + g(x-0) = F(x) + g(x) = \varphi(x), & \text{by (3).} \\ u_t(x,0) = F'(x+0) - g'(x-0) = F'(x) - g'(x) = \psi(x), & \text{by (4)} \end{cases}$$

That is

(7) $$F(x) + g(x) = \varphi(x)$$

(8) $$F'(x) - g'(x) = \psi(x)$$

(7) + (8) would be just solving a system of 2 eq-ns for unknown f-ns $F(x)$ & $g(x)$ provided we could find $F(x)$ & $g(x)$ out of (8). To do that

(8) $$F'(x) - g'(x) = \psi(x) \quad \Big\} \int_0^x (\cdot)\, d\tau \quad \text{of BS of (8):}$$

$$\int_0^x F'(\tau)\, d\tau - \int_0^x g'(\tau)\, d\tau = \int_0^x \psi(\tau)\, d\tau = \quad \text{by FToC}$$

$$F(x) - F(0) - g(x) + g(0) = \int_0^x \psi(\tau)\, d\tau \quad \Rightarrow$$

$$F(x) - g(x) = \underbrace{F(0) - g(0)}_{C} + \int_0^x \psi(\tau)\, d\tau \quad \Rightarrow$$

(9) $$\boxed{F(x) - g(x) = C + \int_0^x \psi(\tau)\, d\tau}$$

With combing (7), we get the ↓ system:

$$\begin{cases} F(x) + g(x) = \varphi(x) & (7) \\ F(x) - g(x) = \int_0^x \psi(\tau)\, d\tau + C & (9) \end{cases} \quad \Rightarrow$$

(10) $$F(x) = \frac{1}{2}\varphi(x) + \frac{1}{2} \cdot \int_0^x \psi(\tau)\, d\tau + \frac{1}{2} C \quad \&$$

(11) $$g(x) = \frac{1}{2}\varphi(x) - \frac{1}{2} \cdot \int_0^x \psi(\tau)\, d\tau - \frac{1}{2} C \quad \Rightarrow$$

$$\begin{cases} F(x+t) = \frac{1}{2}\varphi(x+t) + \frac{1}{2}\int_0^{x+t} \psi(\tau)\, d\tau + \frac{1}{2} C \quad \& \\ g(x-t) = \frac{1}{2}\varphi(x-t) - \frac{1}{2}\int_0^{x-t} \psi(\tau)\, d\tau - \frac{1}{2} C \end{cases}$$

Then by (2), particular sol-n is

$$u(x,t) = F(x+t) + g(x-t) =$$

$$= \frac{1}{2}\varphi(x+t) + \frac{1}{2}\cdot\int_0^{x+t}\psi(\tau)\,d\tau + \cancel{\frac{1}{2}C} +$$

$$+ \frac{1}{2}\varphi(x-t) - \frac{1}{2}\int_0^{x-t}\psi(\tau)\,d\tau - \cancel{\frac{1}{2}C} =$$

$$= \frac{\varphi(x+t)+\varphi(x-t)}{2} + \frac{1}{2}\left[\int_0^{x+t}\psi(\tau)\,d\tau - \int_0^{x-t}\psi(\tau)\,d\tau\right] = \left[\begin{array}{l}\text{Note}\\ \int_0^{x-t}\psi\,d\tau = \\ = -\int_{x-t}^{0}\psi\,d\tau\end{array}\right]$$

$$= \frac{\varphi(x+t)+\varphi(x-t)}{2} + \frac{1}{2}\left[\int_0^{x+t}\psi(\tau)\,d\tau + \int_{x-t}^{0}\psi(\tau)\,d\tau\right] =$$

$$= \frac{\varphi(x+t)+\varphi(x-t)}{2} + \frac{1}{2}\int_{x-t}^{x+t}\psi(\tau)\,d\tau,$$

giving the desired result. $\blacksquare$

(b) Verifying the sol-n (5).

First let me compute the partials:

(13) $$\frac{\partial u}{\partial x} = \frac{\varphi'(x+t)+\varphi'(x-t)}{2} + \frac{1}{2}\cdot\frac{\partial}{\partial x}\left[\int_{x-t}^{x+t}\psi(\tau)\,d\tau\right] =$$ using Leibniz's Rule.

$$= \frac{\varphi'(x+t)+\varphi'(x-t)}{2} + \frac{1}{2}\left[\int_{x-t}^{x+t}\overbrace{\frac{\partial}{\partial x}\left[\psi(\tau)\right]}^{0}d\tau + \psi(x+t)\cdot\frac{\partial}{\partial x}(x+t) - \psi(x-t)\frac{\partial}{\partial x}(x-t)\right]$$

$$= \frac{\varphi'(x+t)+\varphi'(x-t)}{2} + \frac{1}{2}\left[0 + \psi(x+t) - \psi(x-t)\right] =$$

$$= \frac{\varphi'(x+t)\oplus\varphi'(x-t)}{2} + \frac{1}{2}\left[\psi(x+t)\ominus\psi(x-t)\right].$$

Similar steps give u_t with the only difference with $\overset{(13)}{\ominus}$ vs $\overset{(14)}{\oplus}$ as:

due to Leibiz Rule

(14) $$u_t = \frac{\partial u}{\partial t} = \frac{\varphi'(x+t) \ominus \varphi'(x-t)}{2} + \frac{1}{2}\left[\psi(x+t) \oplus \psi(x-t)\right]$$

&

(15) $$u_{xx} = \frac{\varphi''(x+t) + \varphi''(x-t)}{2} + \frac{1}{2}\left[\psi'(x+t) - \psi'(x-t)\right]$$ &

(16) $$u_{tt} = \frac{\varphi''(x+t) + \varphi''(x-t)}{2} + \frac{1}{2}\left[\psi'(x+t) - \psi'(x-t)\right].$$

Then the IVP

Eq-n (1): $u_{xx} - u_{tt} =$

$$\frac{\varphi''(x+t) + \varphi''(x-t)}{2} + \frac{1}{2}\left[\psi'(x+t) - \psi'(x-t)\right] -$$

$$- \frac{\varphi''(x+t) + \varphi''(x-t)}{2} + \frac{1}{2}\left[\psi'(x+t) - \psi'(x-t)\right] \equiv 0 \checkmark$$

IV1, (3): $u(x,0) =$

$$= \frac{\varphi(x+0) + \varphi(x-0)}{2} + \frac{1}{2}\int_{x-0}^{x+0} \psi(\tau)\, d\tau = \frac{2\varphi(x)}{2} + \frac{1}{2}\overbrace{\int_x^x \psi\, d\tau}^{=0} \equiv \varphi(x) \checkmark$$

IV2, (4): $u_t(x,0) =$ (by 14)

$$= \frac{\varphi'(x+0) - \varphi'(x-0)}{2} + \frac{1}{2}\left[\psi(x+0) + \psi(x-0)\right] =$$

$$= \frac{\varphi'(x) - \varphi'(x)}{2} + \frac{1}{2}\left[\psi(x) + \psi(x)\right] = 0 + \frac{1}{2}\cdot\left[2\psi(x)\right] \equiv \psi(x) \checkmark.$$

Hence the given sol-n in (5) satisfies the IVP. ∎